ToolCUA: Towards Optimal GUI-Tool Path Orchestration for Computer Use Agents

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Paper version: arXiv 2605.12481v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *they fail to determine when to continue with GUI actions and when to switch to tools, leading to suboptimal execution paths*

Adding tools to strong models can hurt: Claude-4.5-Sonnet 61.9% GUI vs 48.4% GUI+tools.

Motivation: the reported gaps

1 **Scarce data, weak signal.** Interleaved GUI-Tool trajectories are scarce; rewards ignore switching quality.

2 **Forked road.** Qwen3VL-8B: 0.01 tool calls; larger models over-call.

Evaluation setup

OSWorld-MCP · 333 feasible · average@3 · 50-step cap · Qwen3-VL-8B base · WindowsAgentArena

- **All online** eval on feasible tasks; no offline step accuracy.
- **Held out:** multi_apps excluded from RL; Windows apps unseen.

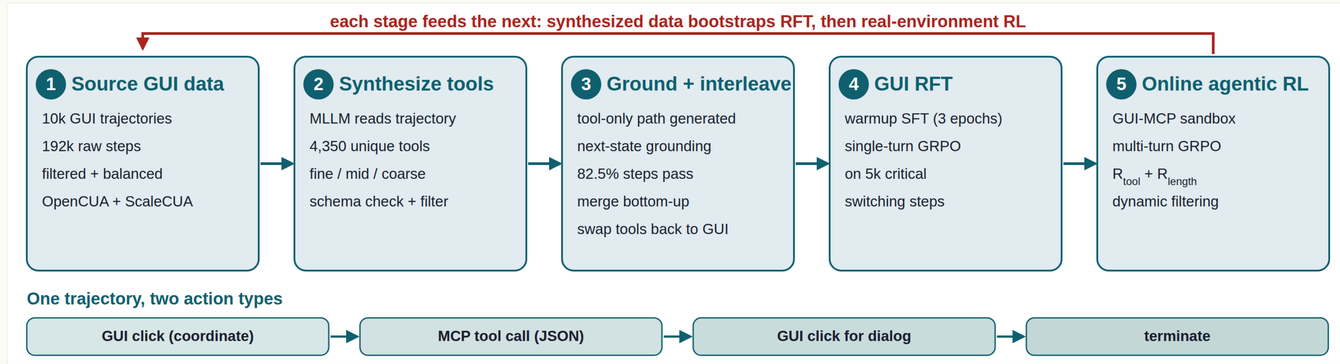
Our read (CUA Radar)

65	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out tasks + platform; baselines; cost; ablations; code.
- **Not reported:** std/CI (average@3 only); matched training compute; grounding-MLLM judge check.
- Pattern C11 (hybrid GUI + tool). Score 65, ★★, transfer.

Method: synthesize the tools, then stage the training

- **Sees / acts:** screenshots + tool results in; coordinate GUI actions or MCP tool calls out.
- **Data:** MLLM synthesizes tools from static GUI trajectories; next-state grounding anchors each simulated call.
- **Signal:** success + format rewards, plus R_{tool} (benefit labels) and R_{length} (group-relative steps).



Our redraw (not the paper's figure): the offline scaling pipeline, the RFT stage, and online RL.

$$R = R_{\text{fmt}} + R_{\text{acc}} + \lambda R_{\text{tool}} + \beta R_{\text{length}}$$

Tool-Efficient Path Reward; λ = 0.4, β = 0.2, shaping active only on success

Key results: OSWorld-MCP accuracy (%)

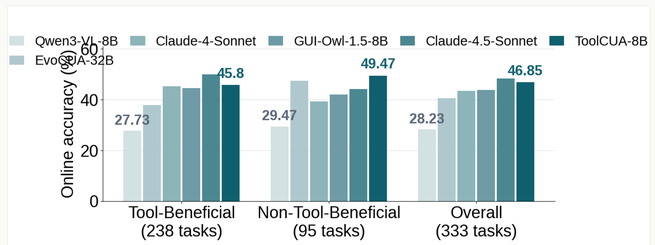


Table 2, redrawn: feasible tasks, average@3, 50-step cap, online.

+18.62 over base (paper's Δ); ACS 14.93 is the lowest reported.

SO WHAT → *TIR 8.41 to 24.32; trails Claude-4.5-Sonnet by 1.50 pts.*

Ablation: training recipe (OSWorld-MCP %)

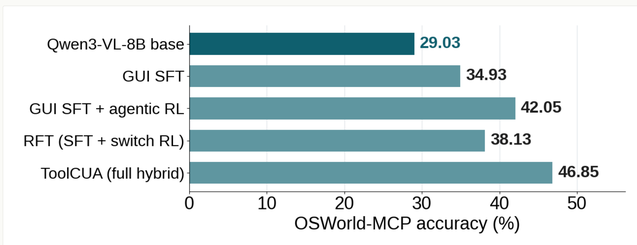


Table 3, redrawn: pure-GUI pipeline vs hybrid GUI-Tool training.

SO WHAT → *Hybrid beats pure-GUI at every stage: RFT 38.13 vs SFT 34.93; full 46.85 vs GUI RL 42.05. Fig. 6: w/o path reward, ~7 pts gap after 20 steps; w/o interleaved data, TIR only ~15%. Held-out multi_apps: 23.9 vs 9.8 base, 18.5 after RFT.*

Headline Numbers

46.85%
OSWorld-MCP accuracy
avg@3; ~66% relative over base

+3.9 pts, GUI-only to tools

23.9% held-out multi_apps

33.8% unseen Windows (WAA)

The paper's own controls

CONTROL	SETTING	RESULT
Held-out tasks	multi_apps excluded	23.9 (base 9.8)
Held-out platform	unseen Windows apps	33.8 WAA GUI-only
Ablation: data	w/o interleaved data	TIR ~15% at end
Ablation: reward	vanilla GRPO	~7 pts gap by step 20

Takeaway

Synthesize the tools, stage the training: 46.85% on OSWorld-MCP with the fewest steps (14.93); gains hold on held-out Linux tasks and Windows.

4,350 synthetic tools · 10k trajs, 180k steps · ACS 19.34 to 14.93 · TIR 8.41 to 24.32 · code released

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